

Engineering Physics: BAPHY105

Linear Operators

Dr. Samir Ranjan Meher

samirmeher@vit.ac.in, samirmeher@gmail.com

Department of Physics, VIT Vellore

Lecture Outcomes

- 1 General definition of an Operator
- 2 Matrix operations
- 3 Eigenvectors and Eigenvalues

Linear Operator

- An operator $\hat{A}$ is a mathematical rule which when applied to a vector, transforms it into another vector in the same vector space.

$$\hat{A}|\psi\rangle = |\psi'\rangle; \quad \langle\phi|\hat{A} = \langle\phi'|$$

- A linear operator preserves the linearity during its operation.

$$\hat{A}(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha\hat{A}|\psi\rangle + \beta\hat{A}|\phi\rangle$$

- In general

✓ Linear operator: $\frac{d}{dx}(\sin x + \cos x) = \frac{d}{dx}\sin x + \frac{d}{dx}\cos x$

✓ Non-linear operator: $\sqrt{(A+B)} \neq \sqrt{A} + \sqrt{B}$

Linear operators as matrices

- ✓ An n -dimensional ket vector, $|\psi\rangle$ can be converted to another n -dimensional ket vector, $|\psi'\rangle$ using an $n \times n$ matrix which is the matrix representation of an operator.

- ✓ In 3 - D Euclidean space, vector $|u\rangle = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ can be projected onto the $x - y$

space by the operator, $\hat{A}$ such that, $\hat{A}|u\rangle = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}.$

✓ Similarly, the **reflection operator** can be represented as,

$$\hat{R}|u\rangle = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}$$

.

✓ The **dilation operator** can be represented as,

$$\hat{D}|u\rangle = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

.

✓ **Rotation operator** causing a rotation of 90° in counter-clockwise direction,

$$\hat{C}|u\rangle = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

.

Few important matrix operations

- **Transpose** of a matrix A^T is obtained by swapping the rows and columns.
- If $A^T = A$ then A is a **symmetric matrix**.
- If $A^T = -A$ then A is an **anti-symmetric matrix**.
- The **adjoint** of a matrix $A^\dagger$ is obtained by taking its conjugated transpose.
- A matrix A is **Hermitian** if $A = A^\dagger$.
- A matrix A is **skew-Hermitian** if $A = -A^\dagger$.
- The **inverse** of matrix A is denoted as A^{-1} such that $AA^{-1} = A^{-1}A = I$. Here I is the **identity matrix**.
- The **commutator** of two matrices A and B is expressed as,

$$[A, B] = AB - BA.$$

Eigenvalue equation

- ✓ For a square matrix A or its equivalent $\hat{A}$, the eigenvalue equation is,

$$\hat{A}|\psi\rangle = \lambda|\psi\rangle$$

- $\lambda \rightarrow$ Eigenvalue
 - $|\psi\rangle \rightarrow$ Eigenvector
- ✓ Eigenvector direction remains unchanged after transformation i.e. the eigenvector is invariant under operator action up to scalar multiplication.
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$$\frac{d}{dx}e^{ax} = ae^{ax}$$

- ✓ The corresponding **characteristic equation** can be expressed as,

$$\det(A - \lambda I) = 0$$

- ✓ The eigenvalues can be determined from the characteristic equation.

Calculation of Eigenvalues & Eigenvectors

- Consider the matrix,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

- The characteristic equation is, $\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 4\lambda + 3 = 0$
- $\Rightarrow \lambda = 1, 3$ i.e. the eigenvalues are 1 and 3.

- Corresponding eigenvectors can be obtained from,

$$(A - 3I)|\psi_1\rangle = 0; \quad (A - I)|\psi_2\rangle = 0$$

$$|\psi_1\rangle \propto \begin{pmatrix} 1 \\ 1 \end{pmatrix}; \quad |\psi_2\rangle \propto \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$